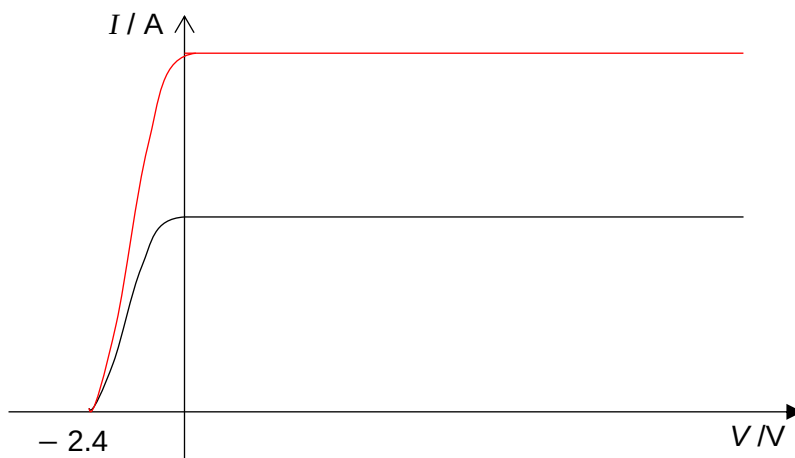


- 4 (a) emission of electrons from a metal surface when surface is irradiated with electromagnetic radiation above threshold frequency B1
- (b) photon has energy dependent on frequency B1
- minimum energy required to remove electron from surface B1
- emission if energy of photon greater than minimum / work function / threshold energy B1
- so, threshold frequency exist
- (c) (i) stopping potential is the minimum potential difference to stop the most energetic electrons from reaching the anode B1
- (ii) for positive values of V , all the electrons emitted from cathode will reach anode B1
- (iii) $E = \phi + eV_s$
 $hf = 1.6 \times 1.6 \times 10^{-19} + 2.4 \times 1.6 \times 10^{-19}$ C1
- $f = 9.7 \times 10^{14} \text{ Hz}$ A1
- (iv) same V_s B1
- saturated current doubled B1



5 (a) (i) magnetic flux density decreases with distance from straight wire so magnetic flux linkage